

David Stone

PO Box 632785, San Diego, CA 92163

Phone: (402) 517-2849

Email: dstone32256@gmail.com

Objective

I would like to bring harmony to the teams and companies with whom I do work.

Employment History

06/2014 - Current **Application Developer (freelance online)**

Google Play, Mountain View, CA

Description -

Development / updates of an app for music generation and other utilities, available for free. The music generation app is a technological advancement for musicians that has 2 algorithms and 5 variations and supports real-time USB-MIDI, BLE-MIDI, and MIDI file generation.

Responsibilities & Accomplishments -

I developed an app to help musicians generate quality music without lots of bulky instruments by acting as a device driver on an Android device controlling MIDI devices and augmenting reality to simulate musicians playing the generated music on actual instruments. The app has become quite popular. I developed 2 other utility apps.

Technologies, Tools, & Methodologies -

Android Studio, Linux, USB-MIDI & BLE-MIDI protocols, SMF / MIDI Files, USB-MIDI Converters, Device Drivers, Real-Time, Generative Music, Algorithmic Music, Android SDK/NDK, C++, Java, JNI, Data Structures, Object Oriented Design/Programming, API usage, Augmented Reality, Multithreading

04/2024 - 10/2024 **Software Developer**

EmployNet (contract for Cubic Defense), San Diego, CA

Description -

I helped to develop embedded software for a communications link. This involved implementation of protocols and sometimes use of open source software.

Responsibilities, & Accomplishments -

I designed the components which I added to the system. I incorporated open source software for the FLUTE protocol into the team's baseline. I implemented the TFTP protocol and incorporated that to the baseline. I helped troubleshoot and debug code and performed various other development tasks.

Technologies, Tools, & Methodologies -

C++, Object-Oriented Design, UML, Socket Programming, UDP, protocols, TFTP, FLUTE, Linux, Semaphores, git, API usage, Embedded Programming, Multithreading

06/2018 - 03/2024 **Software Developer**

SmallBoard.com / OSS Eperts (contract for ASML-Cymer), San Diego, CA

Description -

Help with requirements gathering, design, development, and testing of device drivers, using C++, python, and shell scripting on Linux and VxWorks. Participation in the Agile process (Scrum meetings,).

Responsibilities & Accomplishments -

I contributed to the design and development of various new and existing device drivers and helped out with various other libraries and utilities. The drivers were deployed on Local Controllers as part of an embedded system to lithography machines for microchip manufacturing in Semiconductor / FAB facilities. Some of the drivers were real-time drivers.

Technologies, Tools & Methodologies -

C++, Object-Oriented Design, UML, Python, GoogleTest, GTest / GMock, Linux, Bash Shell Scripting, VxWorks, RTOS, real-time, Object Request Broker (ORB) technology, OPCUA, Siemens, XML, FPGA, SVN, git, API creation and usage, Embedded Programming, FAB, Semiconductor, multitasking, motion control

09/2009 - 06/2014 **Software Developer**

ProKarma (contract for Union Pacific), Omaha, NE

Description -

Help with Requirements gathering, design, development, testing, & support of the company's software.

Responsibilities & Accomplishments -

I performed various enhancements & maintenance on the company's systems for the coordination/planning of intermodal shipments throughout the country. This was a 3-tier application using MFC, C++, C#, .NET framework, ASP.NET, Oracle, ODBC, sockets and EDI. My contribution was heavily the capacity planning and scheduling (Dray, Equipment, & Gate), design and implementation of algorithms (greedy algorithms, DFS,).

Technologies, Tools & Methodologies -

MFC, C++, C#, .NET Framework, ASP.NET, Oracle, SQL, ODBC, EDI (204, 214, 322, 404,), sockets, TCP, Visual Studio on Windows, Object-Oriented Design, UML, data structures, templates, multithreaded programming, database, JavaScript, HTTP protocol, XML, web services, API creation and usage

10/2008 - 03/2009 **Software Developer**

RCS, Ogallala, NE

Description -

Design & develop company software products in Agile / scrum environment.

Responsibilities & Accomplishments -

Help with Software design and development of the company's software products to enable digital recording of spots for radio. I helped with a CD ripper and utilities using Visual C++, .NET Framework, WPF, C#, and MFC.

Technologies, Tools, & Methodologies -

Visual C++, .NET Framework, WPF, C#, MFC, FFT, Visual Studio on Windows, Structural and Object Oriented Programming, data structures, design patterns, API usage

04/2007 - 09/2008 **Software Developer**
ProPrint, Austin, TX

Description -

Research / develop / prototype for the company's business processes using JDF.

Responsibilities & Accomplishments -

Help with a prototype website with back-end JDF technologies. This included PuzzleFlow and PDF JobReady. Back-end integration was accomplished using Microsoft Visual Studio with C#, .NET Framework, and asp.net. CGI programs were developed using C++.

Technologies, Tools, & Methodologies -

Microsoft ASP.NET, Microsoft Visual C#, .NET Framework, Microsoft Visual C++, XML, DOM & SAX, Windows, JDF, JMF, PuzzleFlow, PDF JobReady, SQL Server, SQL, Stored Procedures, Microsoft Windows 2003 Server, Web GUI Development, Middleware Development, Multithreading / Multiprocessing, Visual Studio on Windows, Structural and Object Oriented Programming, Data Structures, Templates, Multithreaded Programming, HTTP Protocol, database, javascript, API usage

10/2004 - 03/2007 **Software Developer**
Universal Software Solutions, Inc, Austin, TX

Description -

Design, develop, support, maintain, and manage the company's software products.

Responsibilities & Accomplishments -

Take responsibility for claims processing for clients using the practice management software, including ANSI, NSF, HCFA, UB92, remittance, patient eligibility, and UK claims. I also developed web-related initiatives used by the customers, including web services and / or html interfaces for drug searches, allergy downloads, drug-drug and drug-allergy interactions, drug monographs, and drug formulary. I was also responsible for developing an HL7 interface, integrating it with existing customer's software and also demonstrating this capability. Development was using Microsoft Visual Studio, Visual C++ and MFC.

Technologies, Tools, & Methodologies -

Microsoft Visual C++, MFC, Windows, SOAP Web Services, XML, DOM & SAX, HTTP, HTML, Relational Database, SQL Server, SQL, Stored Procedures, OOD & documentation, MFC GUI development, Middleware development, Thick Client development, Multithreading / Multiprocessing, Visual Studio on Windows, Structural and Object Oriented Programming, data structures, templates, multithreaded programming, database, API usage

02/2000 - 10/2004 **Software Developer**
Verizon Communications (contract to hire), Irving, TX

Description -

Design, develop, support, maintain, and manage the company's information systems specifically back-end development.

Responsibilities & Accomplishments -

Help with the company's LNP ("Local Number Portability") and various website projects. Mostly, this was middleware and back-end development. Also, I helped with development and testing of the company's data mining efforts. LNP development was accomplished on a Unix platform using C++ and MQ-Series. Backend web development was accomplished with Microsoft Visual Studio using C# and .NET Framework.

Technologies, Tools, & Methodologies -

AIX, C++, MQ-Series, Windows, HTTP, SOAP Web Services, XML, DOM & SAX, C#, Windows Component Services, POP3, SMTP, Relational Database, SQL Server, DB2, SQL, Stored Procedures, Data Mining, Object-Oriented Design, UML, COM, COM+, Middleware Development, Multithreading / Multiprocessing, Visual Studio on Windows, Unix, data structures, design patterns, multithreading, API creation and usage

07/1995 - 02/2000 **Consultant**

Booz-Allen & Hamilton, Mc Lean, VA

Description -

Design and develop applications for the company's clients.

Responsibilities & Accomplishments -

Develop middleware / back-end applications

Technologies, Tools, & Methodologies -

VAX/VMS, Solaris, Windows, C++, Java, EJB, Socket Programming, TCP, Internet Protocols socket implementation, HTTP, HTML, SMTP, POP3, IMAP, FTP, Database Interface, Oracle, Relational Database, Object Oriented Database, Object Store, CORBA, Object Request Broker (ORB), SQL, Stored Procedures, OOD & documentation, X-Windows Motif GUID development, XML, DOM & SAX, Middleware development, Thick Client development, Multithreading / Multiprocessing, Structural and Object Oriented Programming, data structures, design patterns, databases, scripting (shell scripting, Perl, etc.), API usage

Education History

Computer Science

Bachelor's Degree

University of Maryland, College Park, MD

I graduated in May 1995. I majored in Computer Science and minored in Music. In my Numerical Analysis and Electronic Music Composition classes, I learned DFT and FFT and their practical application in digital audio signal processing (DSP). Also, my required math studies included Calculus, Linear Algebra, and Statistics and Probability; and I was required to complete courses in Discrete Mathematics.

Computer Science areas of concentration:

- Algorithms
- Data Structures
- Computer Organization and Architecture
- Numerical Analysis
- Artificial Intelligence
- Elements of Programming Languages

Music areas of concentration:

- Electronic Music Composition
- Jazz Theory

Additional Information

For more information, please see my website:

<https://sites.google.com/view/enlightenmentconcepts>